IEDE Autumn 2026

AI-Assisted Resilience for SDN-Enabled Intelligent Transportation Systems

Trinh Hoang Tu — HUFLIT; Software Engineering Intern, Grab

Advisor: Prof. Kris Singh

Research prototype / evidence console

Motivation: Resilient ITS/V2X

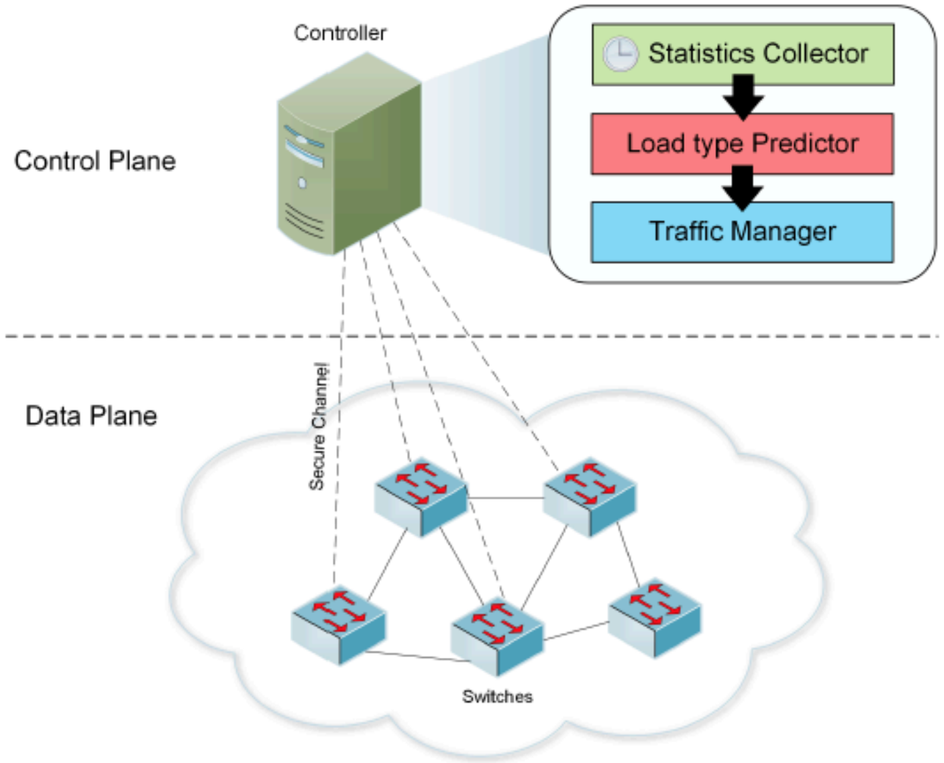


- ITS/V2X services require low latency and continuity.
- SDN simplifies coordination but makes the controller a potential choke point.
- DDoS, PacketIn floods, and misconfiguration can disrupt transport services.
- Permanent blocks may harm legitimate rush-hour traffic and safety.
- Need lightweight detection, reversible containment, and camera context as support only.

Problem Statement

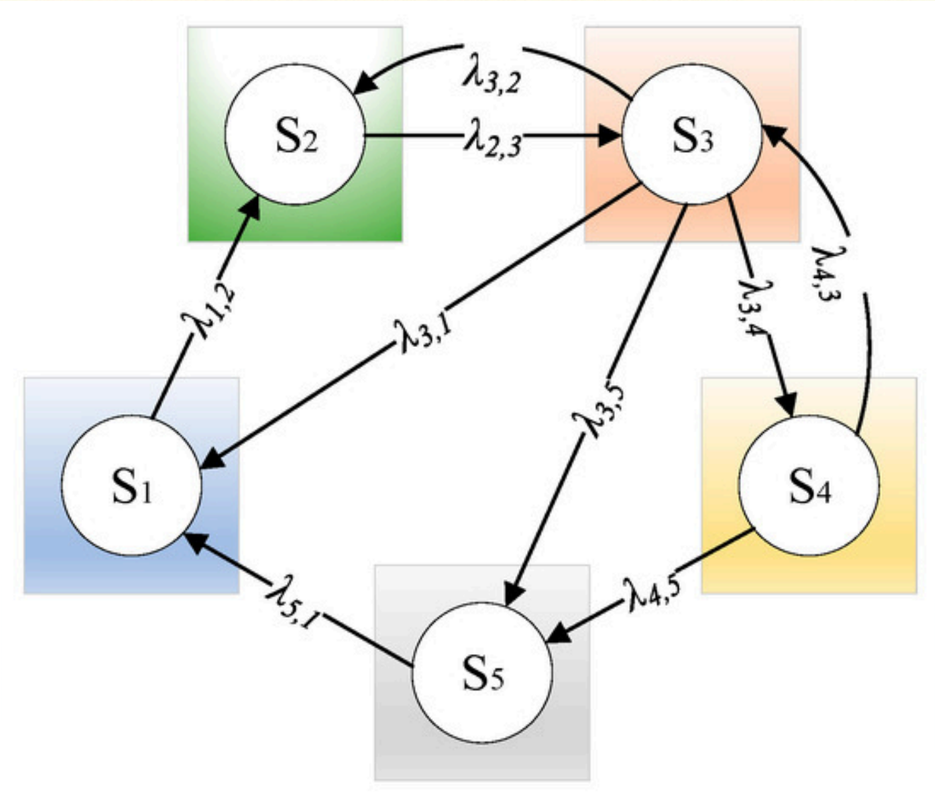
- Detect anomalies from flow-level telemetry without DPI.
- Meet edge latency constraints with a lightweight temporal model.
- Contain suspicious flows without permanent blackholing.
- Verify rollback through graded, auditable evidence.

The research challenge is reliable action under incomplete evidence.



Flow telemetry → TCN-GRU → Adaptive τ → Rollback → Evidence

From Telemetry to Safe Action



Telemetry → TCN-GRU → adaptive τ

temporary OpenFlow → rollback → optional camera fusion

Flow window · Temporal classifier · Threshold

Temporary contain · Cooldown recovery · Physical context

Research → Prototype → Graded Evidence → Extension

Research Contributions

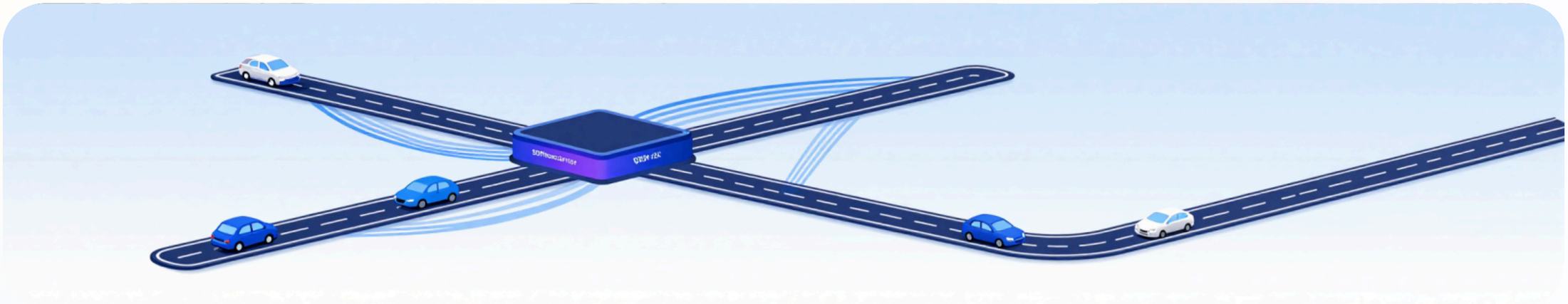


- Non-DPI anomaly detection from SDN flow telemetry.
- Lightweight TCN-GRU inference for edge-oriented response.
- Adaptive-threshold policy for temporary containment.
- Rollback-first mitigation preserves legitimate transport services.
- Camera-assisted fusion provides explainable physical context.

Engineering contribution: evidence console separates MODEL, OF, and NET·EVIDENCE claims.

Threat Model and Scope

- Trusted but overloaded controller: pressure may delay decisions.
- RSU/access segment may generate abnormal PacketIn or flow bursts.
- Camera context may be stale, incomplete, or unavailable.
- Camera evidence must support, not override, network evidence.



Controller pressure

RSU flow bursts

Stale camera context

Out of scope: full controller compromise and adversarial video manipulation.

Evidence Grades: Claim Only What Is Verified

model output → scripted validation → live observation → mock actuation → OpenFlow lab evidence

MODEL output



Scripted validation



OpenFlow lab evidence



We only claim what the grade allows.

Preliminary Results and Honest Boundaries

- InSDN classifier head: Acc ≈ 96.2%, F1 ≈ 97.7%, AUC ≈ 0.977.
- Tracked response: T_response < 10 ms.
- Experiment A: temporary DROP + DELETE rollback; ≈ 5 ms rule recovery.
- CIC zero-shot transfer is weak; adaptation materially improves the result.
- Camera: scripted policy validation, not labeled-video false-positive proof.







Preliminary evaluation — not production validation.

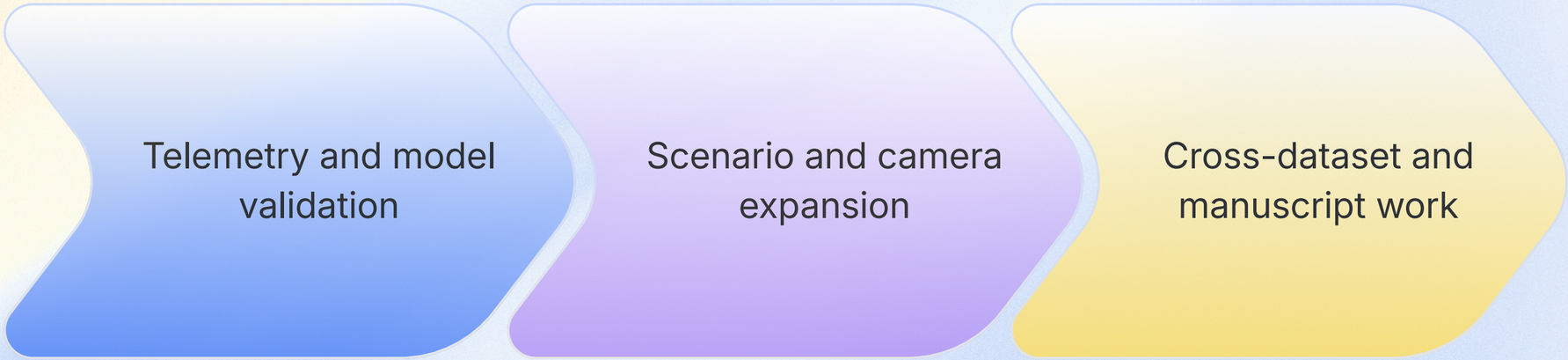
- Multi-seed reproducibility
- Closed-loop experiments
- Artifact release

Next Phase: Work Packages

1. Multi-seed reproducibility
2. Ryu PacketIn evaluation
3. SUMO/VANET scenario expansion
4. Labeled camera data
5. Cross-dataset validation
6. Closed-loop network experiments
7. Manuscript and artifact release



Research prototype → reproducible evidence → broader validation



Thank You

Github repo: thtcsec/sdn-its-resilience-ai

Author: Trinh Hoang Tu · Advisor: Prof. Kris Singh

